SOUMIT PANDEY

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Summary: Researcher with a master's in economics. Experienced in economic modelling, primary survey, qualitative & quantitative primary & secondary research, cross-sectional & longitudinal data econometric analysis, and proficient in environment & sustainability assessment. Skilled in research & academic tools like STATA, R & Game Theory. Dedicated to research with strong analytical, teamwork, comprehension, and communication skills.

EDUCATION

Institute/College/University	Degree	CGPA (*out of 10*)
TERI School of Advanced Studies, New Delhi *(September 2022 - June 2024)*	M.Sc. Economics (Specialization in Environment & Resource Economics)	CGPA 8
Ramakrishna Mission Vidyamandira, University of Calcutta *(July 2019 – June 2022)*	B.Sc. Economics	CGPA 8.8

RESEARCH EXPERIENCES

Role	Organization	Research work	Method
Junior Research Associate	Ashoka Centre for People Centric Energy Transition (ACPET) *(July 2024 – Present)*	- Demand and Supply Assessment & Circular Economy of Critical Minerals of India. - Impact of Clean Cooking Fuels on Health and Livelihoods of rural women in Jharkhand. - Behavioral Nudges to promote Sustainable Mobility	- Systematic Literature Review - Secondary data Quantitative Analysis on Excel - Meta Analysis - Energy Modelling - Secondary data Qualitative Analysis on Excel - Bibliometric Analysis on VoSViewer.
Research Intern	Aseries Envirotek India Pvt. Ltd. *(June 2023 - August 2023)*	- Development of Professional & Academic Courses on Sustainable Development: Prepared modules, white papers, and presentations on sustainable development, ESG, and GRI Reporting, for the purpose of skill development training.	- Systematic Literature Review - Case Study Analysis - Meta Analysis - Quantitative Analysis on Excel -
Subject Matter Expert	Chegg Inc *(September 2021 - February 2023)*	- Subject Matter Expert in Economics: Helped students solve problems based on Economics, Statistics, & Econometric applications.	- Fundamental and Advanced subject knowledge dissemination.

RESEARCH PROJECTS

Output & Project Title	Research Work	Method
Report, paper, article & op-eds: Critical Mineral Requirement Assessment for Clean Energy Technologies in India: A 2025-2070 Net Zero Economy Outlook *(September 2024 – October 2025)*	- Demand & supply assessment of critical minerals required in clean energy technologies. - Material intensity assessment of 30 critical minerals essential to these renewable energy technologies and market share analysis and projection of these technologies. - Projecting demand of the critical minerals till 2070 under Business-as-usual and Net-Zero scenarios.	- Systematic Literature Review - Meta Analysis using Excel & R - Demand Modelling using Excel
Report & Paper: Optimization of India's	- Estimated the determinants of India's critical mineral imports using an econometric framework.	- Meta Analysis of UN Comtrade import data.

Import Portfolio of critical minerals using a mixed effects & PPML model (*July 2025 – April 2026)*	- Constructed counterfactual, model-implied rankings of optimal import partners across ores, products, and scrap categories. - Developed a forward-looking, optimized import portfolio that informs strategic sourcing decisions.	- Supply & trade vulnerability assessment using STATA
Paper: Identifying the Importance of the Incentives and Finding Barriers of Vehicle Scrappage Policy in India *(September 2025 - November 2025)*	- Analysis of Vehicle Scrappage Policy of India and other successful case studies around the world. - Primary survey of expert opinions - Quantitative and Qualitative analysis	- Systematic Literature Review - Stakeholder Analysis - MCDM Analysis (AHP) - Frequency Analysis
Paper: Impact of ESG performance on AI adoption: Case Study of MSMEs of Faridabad Through an Econometric Modelling Approach *(February 2025 – present)*	- Proxy variables to construct ESG and AI variables. - Multiple Imputation Chained Equations (MICE) to overcome missing data challenge - 2SLS/IV estimation to solve endogeneity	- Systematic Literature Review & Meta Analysis to construct proxies. - MICE & 2SLS/IV estimation using STATA
Paper: Socio-economic Viability of Nature-Based Solutions: A Case Study in Kolkata *(February 2024 – May 2025)*	- Contingent Valuation Survey (CVM) to study stated preference of households of the study areas. - Ex-ante Social Cost-Benefit Analysis (CBA) to study the socio-economic prospect of the proposed NBS for a 25-year period (2025-205). - Principal Component Analysis (PCA) to reduce dimensionality of Likert scale variables. - Bi-Variate Probit Model to study the willingness to pay (WTP) scenarios of the CVM study.	- Stakeholder Analysis - Systematic Literature Review - Case study analysis - Preliminary data analysis of survey data, meta-analysis & CBA on Excel - Bi-Variate Probit model & PCA on STATA

PUBLICATIONS

- **Report:** NITI Aayog. (2026, February). Scenarios towards Viksit Bharat and Net Zero: Critical Mineral Assessment: Demand and Supply (Vol. 10). Government of India. https://niti.gov.in/sites/default/files/2026-02/Scenarios-Towards-Viksit-Bharat-and-Net-Zero-Critical-Mineral-Assessment-Demand-and-Supply.pdf
- **Research Paper:** Pandey, S., Chaudhuri, R. R., & Das, S. (2025). Socio-economic viability of nature-based solutions: a case study in Kolkata. *Discover Environment*, *3*(1), 1-20. https://rdcu.be/eoklY
- **Research Paper:** Chaudhary, V., Goswami, A., & Pandey, S. (2025). Impact of AI on ESG performance of firms of Haryana. *Journal of Informatics Education and Research, 5*(3). https://doi.org/10.52783/jier.v5i3.3227
- **Research Article:** Ghosh, A., & Pandey, S. (2025). Resource reality: Mapping India's critical mineral needs for clean energy success. *Terra Green Earth Matters*, 17(12), 14-19. wsds.teriin.org

OPINION PIECES

- Pandey, S. & Ghosh, A. (2026, June 25). *India's Critical Mineral Challenge in a New Global Resource Order*. The Diplomat. https://thediplomat.com/2026/06/indias-critical-mineral-challenge-in-a-new-global-resource-order/
- Ghosh, A., Pandey, S., & Chowdhary, V. (2025, March 3). *Securing critical minerals: A must for India's clean energy ambitions*. 360info. [Republished by Eco-Business, NE News, Big News Network, Kashmir Times]. https://360info.org/securing-critical-minerals-a-must-for-indias-clean-energy-ambitions/
- Pandey, S., Ray Chaudhuri, R., & Das, S. (2025, July). *Nature-based solutions can help Kolkata tackle urban flooding*. *The Indian Express*. https://indianexpress.com/article/opinion/columns/flood-in-kolkata-how-to-tackle-flood-flood-in-himachal-10120577/
- Chaudhary, V., Goswami, A., & Pandey, S. (2025, December 2). *AI adoption in MSMEs of India: Are we ready to face the future?* Postman News. https://postmannews.com/ai-adoption-in-msmes-of-india-are-we-ready-to-face-the-future/

WORKING PAPERS

- Chaudhary, V., Pandey, S. & Goswami, A. (2026). Impact of ESG performance on AI adoption: Case Study of MSMEs Through an Econometric Modelling Approach.
- Ghosh, A., Pandey, S., Chowdhary, V., & Adhikari, A. (2026). Critical mineral requirement assessment for India's net-zero transition: Demand pathways for clean energy technologies.

- Pandey, S. & Ghosh, A. (2026). Optimizing India's Critical Mineral Import Portfolio: A Forward-Looking Mineral-Wise Assessment of Global Sourcing Partners.
- Ghosh, A., Bhattacharjee, D., Pandey, S., & Chowdhary, V. (2026). Identifying the Importance of the Incentives and Finding Barriers of Vehicle Scrappage Policy in India.

RESEARCH PROPOSALS

- Disrupting Strategic Discrimination in Labor Markets: A Framework for Market Segmentation, Technological Change, and Inclusive Equilibria.
- Labour Market Impacts of Urban Flooding and Climate Adaptation: Analysis of Nature-Based Solutions.

INVITED PRESENTATIONS

- **6th November 2025** at **2nd European Congress on Recycling and Waste Management, Vienna, Austria** on "Strategies to Address India's Demand & Supply of Critical Minerals".
- **14th December 2025** at **POMS International Conference, IIM Sambalpur, India** on "Identifying the Importance of the Incentives and Finding Barriers of Vehicle Scrappage Policy in India".
- **16th December 2025** at **International Conference on Sustainable Materials and Infrastructure, IIT Jammu, India** on "Identifying the Importance of the Incentives and Finding Barriers of Vehicle Scrappage Policy in India".

FELLOWSHIPS

Receiver of Hemendra Kothari Fellowship Grant 2024 from TERI School of Advanced Studies *(February 2024)*.

AWARDS

Selected to propose an original research idea to LiFE, NITI Aayog: *"Strategies to Reduce the Flow of Plastic Wastes from FMCG Products" (January 2023 – April 2023)*: Developed a behavioural economics-based proposal focusing on controlling the demand side of FMCG waste generation through institutional interventions.

RELEVANT COURSEWORKS

• Mathematical Economics (Calculus, Linear Algebra, Real Analysis, & Optimization)	• Advanced Econometrics	• Methods of research in economics	• Auctions and Mechanism Design
• Probability and Statistics	• Growth Economics	• Development Economics	• Political Economy
• Microeconomics	• Game Theory • Resource Economics	• Trade, Development & Environment	• Financial Economics
• Macroeconomics	• Environmental Economics	• Theory of Contract	• Public Economics

SKILLS

- Quantitative & Qualitative data analysis: *R Programming, STATA, Excel*
- Bibliometric Analysis: *VoSViewer*
- Primary Survey: *Sampling Strategy, Questionnaire Design, Stakeholder Analysis, FGD, Kobotoolbox, SurveyCTO*
- Optimization Techniques
- Game Theory: *GAMS*
- Meta Analysis & Systematic Literature Review

REFERENCES

Name	Position	Relationship	Email Id
Prof. Sukanya Das	Dean (Research & Partnership)	Professor & master's Thesis Supervisor	sukanya.das@terisas.ac.in
Dr. Sanyyam Khurana	Assistant Professor	Professor	sanyyam.khurana@ashoka.edu.in
Dr. Priyanka Arora	Assistant Professor	Professor	priyanka.arora@terisas.ac.in

EXTRACURICULAR ACTIVITIES

- Co-founder and core member of Oikonomia, the Economics Club of TERI SAS. (September 2023 - May 2024)
- Editor and core member of Vasundhara, a magazine of TERI School of Advanced Studies to address the issues like climate change, energy crisis, environmental crisis, marginalized communities, etc. (June 2023 - May 2024)
- Vice President of Ramakrishna Mission Vidyamandira Photography Club. (June 2021 - June 2022)